01. Business Problem & Objective

A multi-stage NLP pipeline for theme detection, brand-level sentiment and article headline generation.

Project: Automated Mobile-Technology Brand Sentiment and Headline Generator

Source archive: NLP-Sentiment-Classification-of-Mobile-brands-main.zip | Notebook: main.ipynb | Supporting report and presentation included

Business context

Digital content such as news articles and social posts is produced at high volume and in multiple languages. The task is to help an analyst quickly identify mobile-technology content, determine which mobile brands are mentioned and what sentiment is expressed toward each brand, and generate a concise English headline for relevant articles.

Problem decomposition

This is not a single sentiment classifier. It is a pipeline of linked NLP tasks with different outputs and evaluation criteria.

Stage	Input → Output	Task type
1. Theme identification	Tweet/article → mobile technology or other	Binary classification
2. Brand identification	Mobile-themed text → list of brand entities	Dictionary/regex entity matching
3. Language processing	Hindi/Hinglish → English	Language detection + translation
4. Brand sentiment	Text + target brand → negative/neutral/positive	Target-dependent classification
5. Headline generation	Mobile-tech article → English headline (≤20 words)	Abstractive text generation

Business objectives

- Reduce manual review time across large article and tweet streams.
- Avoid assigning sentiment to the whole document when multiple brands have different opinions.
- Produce consistent English outputs from English, Hindi and Hinglish input.
- Generate short, informative headlines for mobile-technology articles.
- Balance quality, inference speed, model size and language scalability.

Success criteria from the challenge

The challenge's composite score is 0.4 × mobile-theme classification F1 + 0.2 × brand-identification accuracy + 0.2 × sentiment F1 + 0.2 × average headline similarity. It also evaluates runtime, innovation and scalability. The repository does not provide all final numeric scores, so do not invent an overall score.

Interview framing

> “I worked on an end-to-end NLP pipeline that identifies mobile-technology content in articles and tweets, extracts mentioned brands, predicts sentiment separately for each brand, and generates English headlines for relevant articles. The key design choice was to use lightweight models for high-volume filtering and more advanced contextual models only for the harder downstream tasks.”